



Full length article

Pan-Mamba: Effective pan-sharpening with state space model[☆]

Xuanhua He^{a,b}, Ke Cao^{a,b}, Jie Zhang^{a,b,*}, Keyu Yan^{a,b}, Yingying Wang^c, Rui Li^a, Chengjun Xie^a, Danfeng Hong^d, Man Zhou^{b,**}

^a Hefei Institutes of Physical Science, Chinese Academy of Sciences, Hefei, 230031, China

^b University of Science and Technology of China, Hefei, 230026, China

^c Key Laboratory of Multimedia Trusted Perception and Efficient Computing, Xiamen University, Xiamen, 361102, China

^d Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, 100094, China

ARTICLE INFO

Keywords:

Pan-sharpening
State-space model
Image fusion

ABSTRACT

Pan-sharpening involves integrating information from low-resolution multi-spectral and high-resolution panchromatic images to generate high-resolution multi-spectral counterparts. While recent advancements in the state space model, particularly the efficient long-range dependency modeling achieved by Mamba, have revolutionized computer vision community, its untapped potential in pan-sharpening motivates our exploration. Our contribution, Pan-Mamba, represents a novel pan-sharpening network that leverages the efficiency of the Mamba model in global information modeling. In Pan-Mamba, we customize two core components: channel swapping Mamba and cross-modal Mamba, strategically designed for efficient cross-modal information exchange and fusion. The former initiates a lightweight cross-modal interaction through the exchange of partial panchromatic and multi-spectral channels, while the latter facilitates the information representation capability by exploiting inherent cross-modal relationships. Through extensive experiments across diverse datasets, our proposed approach surpasses state-of-the-art methods, showcasing superior fusion results in pan-sharpening. To the best of our knowledge, this work is the first attempt in exploring the potential of the Mamba model and establishes a new frontier in the pan-sharpening techniques. The source code is available at <https://github.com/alexhe101/Pan-Mamba>.

1. Introduction

With the rapid development of satellite technology, there exists a great need for high-resolution multi-spectral (HRMS) remote sensing images across various fields. However, owing to constraints imposed by physical sensors, acquiring high-resolution multi-spectral images directly through remote sensing satellites poses a considerable challenge. Therefore, satellites are equipped with two distinct types of panchromatic (PAN) and multi-spectral sensors, designed to capture low-resolution multi-spectral (LRMS) images and high-resolution texture-rich PAN images. These two sets of images with complementary information are subsequently fused by pan-sharpening technique, which enables the acquisition of high-resolution multi-spectral images. In recent years, pan-sharpening technology has garnered considerable attention, with methods broadly categorized into classical approaches and deep learning-based techniques. Classical methods encompass component substitution methods, multi-resolution analysis methods, and

variational optimization (VO) based methods. These traditional pan-sharpening approaches have predominantly relied on manually designed physical rules [1–6]. However, they often faced challenges stemming from insufficient representation capabilities, ultimately leading to suboptimal results. Inspired by the powerful learning capability of deep learning, the pioneer PNN [7] marked a pivotal moment by integrating deep learning methods into this domain, showcasing great advancements over traditional techniques. Subsequently, a flood of increasingly complex deep learning-based models has been introduced, encompassing multi-scale methods [8], models leveraging frequency domain information [9,10], architectures based on Transformer models [11], models incorporating flow model [12], and those driven by domain-specific prior knowledge [13].

Nevertheless, the aforementioned promising methods exhibit certain limitations that hinder further performance improvement. Specifically, the first challenge lies in capturing the global information.

[☆] This work was supported by the Anhui Provincial Natural Science Foundation, Grant No. 2408085MD090.

* Correspondence to: 350 Shushanhu Road, Hefei 230031, Anhui, China

** Corresponding author.

E-mail addresses: zhangjie@iim.ac.cn (J. Zhang), manman@mail.ustc.edu.cn (M. Zhou).

INNformer [14] and Panformer [11] attempt to model global information by incorporating vision transformer [15] and Swin Transformer blocks [16], respectively. However, the former introduces computational complexity, rendering its application challenging, while the latter's window partitioning imposes constraints on the model's receptive field and disrupts the locality of features [17]. SFNet [9] and MSDDN [10] adopt a different approach by introducing Fourier transform to model global information. However, the interaction between the frequency domain and spatial domain introduces information gaps [18], and the fixed convolution parameters hinder the model's adaptive ability to varying inputs. Conversely, some techniques that aim to reduce the complexity of self-attention, such as window partitioning [16] and transposed self-attention [19], sacrifice to some extent the inherent capabilities of self-attention, including global information modeling and input adaptability. Secondly, current methods exhibit shortcomings in adequate and efficient information fusion. Panformer employs window-based cross attention, INNformer and PanFlowNet rely on invertible neural networks, and SFNet utilizes a Fourier convolution for feature fusion. However, these approaches fall short in fully capturing the cross-modal correlations between the two modalities and suffer from high computational complexity. The birth of the Mamba [20] offers a novel solution to the aforementioned challenges. It features input-adaptive and global information modeling capabilities akin to self-attention, while maintaining linear complexity, reduced computational overhead, and enhanced inference speed. Notably, in the realm of natural language processing, the Mamba model has demonstrated superior results compared to the Transformer architecture.

Considering these considerations, our approach focuses on enhancing models through two key perspectives: feature extraction and feature fusion. We introduce Pan-Mamba, a pan-sharpening network that leverages Mamba as the core module. Our design includes channel swapping Mamba and cross modal Mamba for global information extraction and efficient feature fusion from both PAN and LRMS images. The channel swapping Mamba initiates a preliminary cross-modal interaction by exchanging partial PAN channels and LRMS channels, facilitating a lightweight and efficient fusion of information. Meanwhile, the cross modal Mamba utilizes the inherent cross-modal relationship between the two, enabling fusion, filtering of redundant modal features, and obtaining refined fusion results. Owing to its efficient feature extraction and fusion capabilities, our model has surpassed state-of-the-art methods, achieving superior fusion results. Our contributions can be summarized as follows:

- This work is the first attempt to introduce the Mamba model into the pan-sharpening domain and presents a novel pan-sharpening network. This approach facilitates efficient long-range information modeling and cross-modal information interaction.
- We tailor the channel-swapping Mamba block and cross-modal Mamba block to enable efficient exchange and fusion of cross-modal information.
- Conducting extensive experiments across multiple datasets, our proposed method demonstrates state-of-the-art results in both qualitative and quantitative assessments.

2. Related work

2.1. Pan-sharpening

The methods of pan-sharpening are primarily categorized into two parts: traditional approaches and deep learning-based methods. Traditional methods predominantly rely on manually designed priors, encompassing component substitution algorithms [1,2,21,22], multi-resolution analysis algorithms [23,24], and variational optimization algorithms [25]. Component substitution algorithms leverage the spatial details of PAN images to replace the spatial information of LRMS images. Multi-resolution methods conduct multi-resolution analysis and

subsequently fuse the two images, whereas variational optimization-based algorithms model the fusion process as an energy function and iteratively solve it. These methods suffer from limitations in performance due to their insufficient feature representation. The rise of deep learning in pan-sharpening was initiated by the PNN [7] model, which, drawing inspiration from SRCNN [26], devised a simple three-layer neural network with promising outcomes. Subsequent advancements introduced more complex designs to this domain, such as PanNet [27] utilizing ResNet blocks to capture high-frequency information, MSD-CNN [8] introducing multi-scale convolution for processing the multi-scale structure of remote sensing images, and SRPPNN [28] employing a progressive upsampling strategy. The emergence of Transformers has influenced pan-sharpening, with INNformers [14] and Panformers [11] introducing self-attention mechanisms. SFNet [9] and MSDDN [10] employ Fourier transforms to capture global features and facilitate the learning of high-frequency information. PanFlowNet [12] incorporates a flow-based model for processing remote sensing images. Furthermore, prior information-driven methods have been integrated into this field, exemplified by MutNet [29] and GPPNN [13], which leverage prior knowledge between modalities to promote image fusion.

2.2. State space model

The concept of the state space model was initially introduced in the S4 [30] model, presenting a distinctive architecture capable of effectively modeling global information in comparison to conventional CNN or Transformer architectures. Building upon S4, the S5 [31] model emerged, strategically reducing complexity to a linear level. The subsequent H3 [32] model further refined and expanded upon this foundation, enabling the model to perform competitively with Transformers in language model tasks. Mamba [20], in turn, introduced an input-adaptive mechanism to enhance the state space model, resulting in higher inference speed, throughput, and overall metrics compared to Transformers of equivalent scale. Its application in NLP, particularly in large language models [33,34], has proven as effective as transformers, showcasing its exceptional sequence modeling capabilities. With Mamba's success in NLP, the community has increasingly adopted it for tasks such as time series analysis [35] and speech processing [36]. The application of the State Space model extended into visual tasks with the introduction of Vision Mamba [37] and Vmaba [38]. These models decompose images into smaller patches, which are then organized sequentially, allowing them to be processed as a sequence. By leveraging Mamba's scan mechanism, the models efficiently traverse these sequences, integrating information across patches. This method capitalizes on Mamba's inherent ability to capture global features across an image, while maintaining linear computational complexity. These adaptations yielded commendable results in classification and segmentation tasks, successfully penetrating fields such as medical image segmentation [39,40]. Notably, the potential of this model in multimodal image fusion remains an area that has not been thoroughly explored.

3. Methods

In this section, we begin by introducing the fundamental knowledge of the state space model. Subsequently, we delve into a detailed exploration of our model, encompassing its architectural framework, module design, and our loss function.

3.1. Preliminaries

The state space sequence model and Mamba draw inspiration from linear systems, with the goal of mapping a one-dimensional function or sequence, denoted as $x(t) \in \mathbf{R}$, to $y(t)$ through the hidden space $h(t) \in \mathbf{R}^N$. In this context, $\mathbf{A} \in \mathbf{R}^{N \times N}$ serves as the evolution parameter,

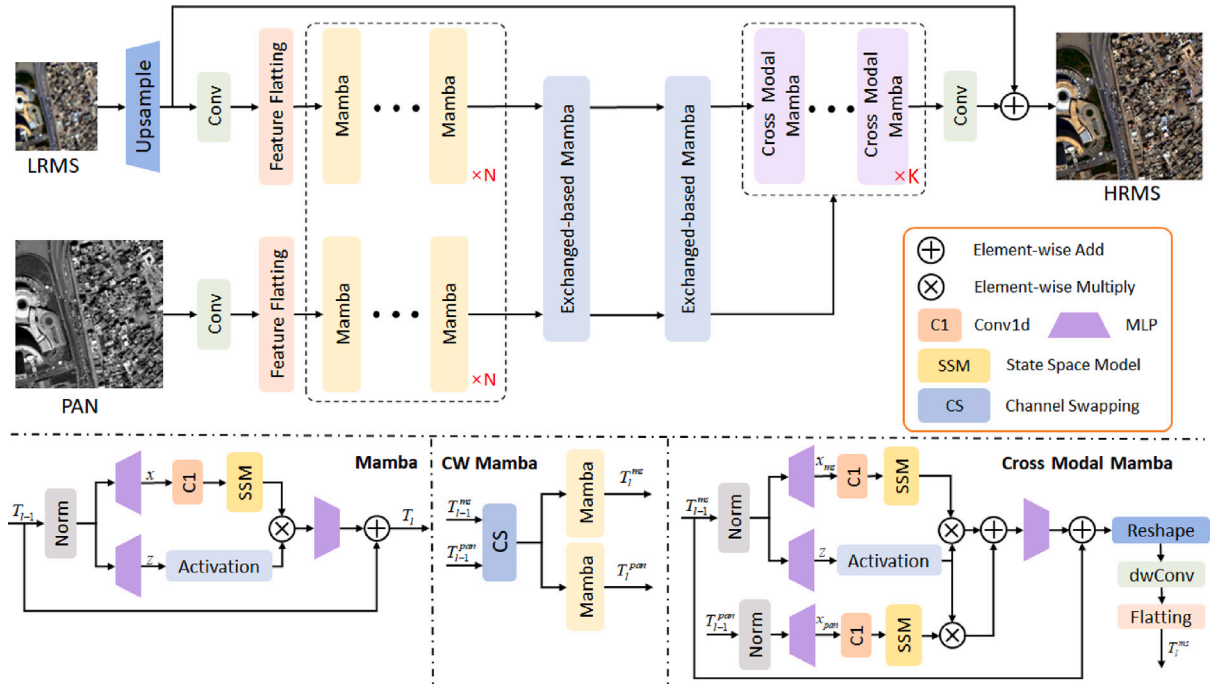


Fig. 1. The network structure of our proposed Pan-Mamba, which includes three Key components: Mamba block for long-range feature extraction, CW Mamba and cross modal mamba for shallow and deep feature fusion.

while $\mathbf{B} \in \mathbf{R}^{N \times 1}$ and $\mathbf{C} \in \mathbf{R}^{1 \times N}$ act as the projection parameters. The system can be mathematically expressed using the following formula.

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad (1)$$

$$y(t) = \mathbf{C}h'(t). \quad (2)$$

The S4 and Mamba models serve as discrete counterparts to the continuous system, incorporating a timescale parameter Δ to convert continuous parameters $\mathbf{A}$ and $\mathbf{B}$ into their discrete counterparts $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$. The prevalent approach employed for this transformation is the zero-order hold (ZOH) method, which can be formally defined as follows:

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \quad (3)$$

$$\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}. \quad (4)$$

The discrete representation of this linear system can be formulated as follows:

$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, \quad (5)$$

$$y_t = \mathbf{C}h_t. \quad (6)$$

Finally, the output is derived through global convolution:

$$\bar{\mathbf{K}} = (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \dots, \mathbf{C}\bar{\mathbf{A}}^{M-1}\bar{\mathbf{B}}), \quad (7)$$

$$\mathbf{y} = \mathbf{x} * \bar{\mathbf{K}}. \quad (8)$$

Here, $\mathbf{M}$ denotes the sequence length of $\mathbf{x}$, and $\bar{\mathbf{K}} \in \mathbf{R}^M$ represents a structured convolutional kernel.

3.2. Network architecture

The architecture of our model is depicted in Fig. 1, comprising three core components: the Mamba block, the channel swapping Mamba block, and the cross modal Mamba block. The Mamba block is instrumental in modeling the long-range dependencies within PAN and LRMS features, while the channel swapping Mamba block and cross-modal Mamba block are employed to explore the relationship between the two

modalities. Given the input LRMS and PAN images denoted as M and P , the network pipeline can be expressed as follows:

Firstly, we employ convolutional layers to project the two images into the feature space and flatten them along the spatial dimension into tokens:

$$\mathbf{M}_f, \mathbf{P}_f = \phi(\mathbf{M}), \phi(\mathbf{P}), \quad (9)$$

$$\mathbf{T}_0^{ms}, \mathbf{T}_0^{pan} = \text{flatten}(\mathbf{M}_f), \text{flatten}(\mathbf{P}_f). \quad (10)$$

Subsequently, $\mathbf{T}_0^{ms}$ and $\mathbf{T}_0^{pan}$ are fed independently into a sequence of Mamba blocks for global feature extraction:

$$\mathbf{T}_i^{ms} = \psi_{m_i} \cdot (\psi_{m_1}(\psi_{m_0}(\mathbf{T}_0^{ms}))), \quad (11)$$

$$\mathbf{T}_i^{pan} = \psi_{p_i} \cdot (\psi_{p_1}(\psi_{p_0}(\mathbf{T}_0^{pan}))). \quad (12)$$

Here, $\psi_{m_i}(\cdot)$ and $\psi_{p_i}(\cdot)$ denote the i th Mamba block for extracting LRMS and PAN features, respectively.

Upon obtaining the global features $\mathbf{T}_i^{ms}$ and $\mathbf{T}_i^{pan}$, we leverage the channel swapping Mamba to enhance feature interaction and obtain $\mathbf{T}_k^{ms}$ and $\mathbf{T}_k^{pan}$. Subsequently, we utilize the cross modal Mamba block for deep feature fusion. After that, we reshape the MS token into the spatial dimension. The final output is obtained through convolution layers and residual connections:

$$\mathbf{T}_{k+j}^{ms} = \theta_j \cdot (\theta_1(\theta_0(\mathbf{T}_k^{ms}, \mathbf{T}_k^{pan}), \mathbf{T}_k^{pan}), \mathbf{T}_k^{pan}), \quad (13)$$

$$\mathbf{M}_{out} = \phi(\text{reshape}(\mathbf{T}_{k+j}^{ms})) + \mathbf{M}. \quad (14)$$

Here, $\theta(\cdot)$ represents the cross modal Mamba block, and $\phi(\cdot)$ denotes the convolution layers for channel adjustment.

3.3. Key components

3.3.1. Mamba block

Motivated by the Mamba, we employ the Mamba block to extract features and model long-range dependencies. A comprehensive overview of the operations is provided in Algorithm 2. Specifically, the input token sequence $\mathbf{T}_{i-1} \in \mathbf{R}^{B \times N \times C}$ undergoes initial normalization via layer normalization. Subsequently, the normalized sequence is projected into $\mathbf{x} \in \mathbf{R}^{B \times N \times P}$ and $\mathbf{z} \in \mathbf{R}^{B \times N \times P}$ using a multi-layer perceptron

(MLP). Following this, a 1-D convolution layer with SiLU activation is applied to process $\mathbf{x}$ and yield $\mathbf{x}'$. Further projection of $\mathbf{x}'$ onto $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ is performed, and the timescale parameter Δ is employed to convert them into discrete versions $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$. The parameter generation process is delineated in Algorithm 1 and corresponds to formula (4). In this algorithm, $(\mathbf{B})$, $(\mathbf{N})$, $(\mathbf{K})$, and $(\mathbf{P})$ represent the batch size, the token count of a feature, the expanded feature channel, and the SSM dimension, respectively. After that, the output $\mathbf{y}$ is computed through the SSM. Subsequently, $\mathbf{y}$ is gated by $\mathbf{z}$ and added to the input $\mathbf{T}_{l-1}$, to get the output sequence $\mathbf{T}_l$. The SSM process is described in formula (8). The computational complexity of Mamba block exhibits linearity to the sequence length N . Specifically, the computational complexity is expressed as $3N(2D)C + N(2D)C$. D and C are fixed numbers.

Algorithm 1 Parameters Function(Function)

Require: token sequence $\mathbf{x}' : (\mathbf{B}, \mathbf{N}, \mathbf{P})$
Ensure: token sequence $\bar{\mathbf{A}} : (\mathbf{B}, \mathbf{N}, \mathbf{P}, \mathbf{K}), \bar{\mathbf{B}} : (\mathbf{B}, \mathbf{N}, \mathbf{P}, \mathbf{K}), \bar{\mathbf{C}} : (\mathbf{B}, \mathbf{N}, \mathbf{P})$

- 1: $\mathbf{B} : (\mathbf{B}, \mathbf{N}, \mathbf{K}) \leftarrow \text{Linear}^{\mathbf{B}}(\mathbf{x}')$
- 2: $\mathbf{C} : (\mathbf{B}, \mathbf{N}, \mathbf{K}) \leftarrow \text{Linear}^{\mathbf{C}}(\mathbf{x}')$
- 3: $\Delta : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \log(1 + \exp(\text{Linear}^{\Delta}(\mathbf{x}') + \text{Parameter}^{\Delta}))$
- 4: $\text{Parameter}^{\Delta} \in \mathbb{R}^{(\mathbf{P}, \mathbf{K})} \text{ */}$
- 5: $\bar{\mathbf{A}} : (\mathbf{B}, \mathbf{N}, \mathbf{P}, \mathbf{K}) \leftarrow \Delta_o \otimes \text{Parameter}^{\Delta}$
- 6: $\bar{\mathbf{B}} : (\mathbf{B}, \mathbf{N}, \mathbf{P}, \mathbf{K}) \leftarrow \Delta_o \otimes \mathbf{B}$

Return: $\bar{\mathbf{A}}, \bar{\mathbf{B}}, \bar{\mathbf{C}}$

Algorithm 2 Mamba Block

Require: token sequence $\mathbf{T}_{l-1} : (\mathbf{B}, \mathbf{N}, \mathbf{C})$
Ensure: token sequence $\mathbf{T}_l : (\mathbf{B}, \mathbf{N}, \mathbf{C})$

- 1: $\text{*/ Apply layer normalization to the input sequence } \mathbf{T}_{l-1} \text{ */}$
- 2: $\mathbf{T}'_{l-1} : (\mathbf{B}, \mathbf{N}, \mathbf{C}) \leftarrow \text{Norm}(\mathbf{T}_{l-1})$
- 3: $\mathbf{x} : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \text{Linear}^{\mathbf{x}}(\mathbf{T}'_{l-1})$
- 4: $\mathbf{z} : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \text{Linear}^{\mathbf{z}}(\mathbf{T}'_{l-1})$
- 5: $\text{*/ process the input sequence */}$
- 6: $\mathbf{x}' : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \text{SiLU}(\text{Conv1d}(\mathbf{x}))$
- 7: $\bar{\mathbf{A}} : (\mathbf{B}, \mathbf{N}, \mathbf{P}, \mathbf{K}), \bar{\mathbf{B}} : (\mathbf{B}, \mathbf{N}, \mathbf{P}, \mathbf{K}), \mathbf{C} : (\mathbf{B}, \mathbf{N}, \mathbf{K}) \leftarrow \text{Function}(\mathbf{x}')$
- 8: $\mathbf{y} : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \text{SSM}(\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C})(\mathbf{x}')$
- 9: $\text{*/ Obtain the gated y by multiplying it with a gating factor. */}$
- 10: $\mathbf{y}' : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \mathbf{y} \odot \text{SiLU}(\mathbf{z})$
- 11: $\text{*/ residual connection */}$
- 12: $\mathbf{T}_l : (\mathbf{B}, \mathbf{N}, \mathbf{C}) \leftarrow \text{Linear}^{\mathbf{T}}(\mathbf{y}') + \mathbf{T}_{l-1}$

Return: $\mathbf{T}_l$

3.3.2. Channel swapping mamba block

To encourage feature interaction between PAN and LRMS modalities and initiate a correlation between them, we introduced a Mamba fusion block based on channel swapping. This module efficiently swaps channels between LRMS and PAN features, facilitating lightweight feature interaction. The swapped features are then processed through the Mamba block. The channel swapping operation enhances cross-modal correlations by incorporating information from distinct channels, thereby enriching the diversity of channel features, which contributes to the overall improvement in model performance. Given LRMS features $\mathbf{T}_{l-1}^{ms} \in \mathbb{R}^{B, N, C}$ and PAN features $\mathbf{T}_{l-1}^{pan} \in \mathbb{R}^{B, N, C}$ as inputs, we split each feature along the channel dimension into two equal portions. The first half of channels from $\mathbf{T}_{l-1}^{ms}$ is concatenated with the latter half of $\mathbf{T}_{l-1}^{pan}$ and processed through the Mamba block for feature extraction. The obtained features are added to $\mathbf{T}_{l-1}^{ms}$, resulting in the creation of a new feature $\mathbf{T}_l^{ms}$. Simultaneously, the first half of $\mathbf{T}_{l-1}^{pan}$ is concatenated with the latter half of $\mathbf{T}_{l-1}^{ms}$ and passed through the Mamba block. The resulting features are added to $\mathbf{T}_{l-1}^{pan}$, generating $\mathbf{T}_l^{pan}$. These features encapsulate information from both modalities, enhancing overall feature diversity.

Algorithm 3 Cross Modal Mamba Block

Require: token sequence $\mathbf{T}_{l-1}^{ms} : (\mathbf{B}, \mathbf{N}, \mathbf{C}), \mathbf{T}_{l-1}^{pan} : (\mathbf{B}, \mathbf{N}, \mathbf{C})$
Ensure: token sequence $\mathbf{T}_l^{ms} : (\mathbf{B}, \mathbf{N}, \mathbf{C})$

- 1: **for** o in $\{\text{pan}, \text{ms}\}$ **do**
- 2: $\text{*/ Apply layer normalization to the input sequence } \mathbf{T}_{l-1} \text{ */}$
- 3: $\mathbf{T}'_{l-1} : (\mathbf{B}, \mathbf{N}, \mathbf{C}) \leftarrow \text{Norm}(\mathbf{T}_{l-1})$
- 4: $\mathbf{x}_o : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \text{Linear}^{\mathbf{x}}(\mathbf{T}'_{l-1})$
- 5: $\text{*/ process with sequence */}$
- 6: $\mathbf{x}'_o : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \text{SiLU}(\text{Conv1d}_o(\mathbf{x}_o))$
- 7: $\bar{\mathbf{A}}_o : (\mathbf{B}, \mathbf{N}, \mathbf{P}, \mathbf{K}), \bar{\mathbf{B}}_o : (\mathbf{B}, \mathbf{N}, \mathbf{P}, \mathbf{K}), \mathbf{C}_o : (\mathbf{B}, \mathbf{N}, \mathbf{K}) \leftarrow \text{Function}_o(\mathbf{x}'_o)$
- 8: $\mathbf{y}_o : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \text{SSM}(\bar{\mathbf{A}}_o, \bar{\mathbf{B}}_o, \mathbf{C}_o)(\mathbf{x}'_o)$
- 9: **end for**
- 10: $\mathbf{z} : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \text{Linear}^{\mathbf{z}}(\mathbf{T}'_{l-1})$
- 11: $\text{*/ Obtain the gated y by multiplying it with a gating factor. */}$
- 12: $\mathbf{y}'_{ms} : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \mathbf{y}_{ms} \odot \text{SiLU}(\mathbf{z})$
- 13: $\mathbf{y}'_{pan} : (\mathbf{B}, \mathbf{N}, \mathbf{P}) \leftarrow \mathbf{y}_{pan} \odot \text{SiLU}(\mathbf{z})$
- 14: $\text{*/ residual connection */}$
- 15: $\mathbf{F}_{l-1}^{ms} : (\mathbf{B}, \mathbf{C}, \mathbf{H}, \mathbf{W}) \leftarrow \text{Reshape}(\text{Linear}^{\mathbf{T}}(\mathbf{y}'_{ms} + \mathbf{y}'_{pan}) + \mathbf{T}_{l-1}^{ms})$
- 16: $\text{*/ depth-wise convolution */}$
- 17: $\mathbf{T}_l^{ms} : (\mathbf{B}, \mathbf{N}, \mathbf{C}) \leftarrow \text{Flatten}(\text{DWConv}(\mathbf{F}_{l-1}^{ms}) + \mathbf{F}_{l-1}^{ms})$

Return: $\mathbf{T}_l^{ms}$

3.3.3. Cross modality Mamba block

Motivated by the concept of Cross Attention [41], we introduce a novel cross modal Mamba block designed for facilitating cross-modal feature interaction and fusion. In this approach, we project features from two modalities into a shared space, employing gating mechanisms to encourage complementary feature learning while suppressing redundant features. Concurrently, to enhance local features, we incorporate depth-wise convolution within the module, thereby amplifying the encoding capacity of local features during the fusion process. The details of this module are elucidated in Algo 3. The generation of $\mathbf{y}_{ms}$ and $\mathbf{y}_{pan}$ follows the process outlined in the Mamba block. Subsequently, we obtain the gating parameters $\mathbf{z}$ by projecting $\mathbf{T}'_{l-1}$ and employ $\mathbf{z}$ to modulate $\mathbf{y}_{ms}$ and $\mathbf{y}_{pan}$. The fusion of these two features involves addition, followed by reshaping to obtain a 2-D feature $\mathbf{F}_{l-1}^{ms}$. To enhance locality, we apply depth-wise convolution, and subsequently flatten the feature to a 1-D sequence, generating output sequence $\mathbf{T}_l^{ms}$.

3.4. Loss function

In alignment with the prevalent practices in this field, we adopt the L1 loss as our chosen loss function [42]. Specifically, with the output denoted as $\mathbf{M}_{\text{out}}$ and the corresponding ground truth as $\mathbf{G}$, the loss function is expressed as:

$$\mathcal{L} = \|\mathbf{G} - \mathbf{M}_{\text{out}}\|_1 \quad (15)$$

4. Experiment

4.1. Datasets and benchmark

In our experiments, we selected datasets comprising WorldView-II (WV2), Gaofen-2 (GF2) and WorldView-III (WV3), characterized by diverse resolutions and a broad spectrum of scenes. Specifically, WorldView-II encompasses industrial areas and natural landscapes, Gaofen-2 encompasses mountains and rivers, while WorldView-III predominantly features urban roads and urban scenes. Given the absence of ground truth, the dataset generating process adheres to the Wald protocol [43]. The details of our datasets are presented in Table 1. Our datasets encompass a wide range of resolutions. For comparative analysis, we opted for a selection of representative traditional methods, including GFPCA [21], GS [44], Brovey [2], IHS [22], SFIM [1], BDSD [3], SFNLR [5] and LRTCFpan [45], alongside advanced deep

Table 1

Detailed information of the datasets used in our experiments.

Dataset information	WV2 dataset	GF2 dataset	WV3 dataset	FGF2 dataset	FWV2 dataset	FWV3 dataset
Training samples	760	2700	2150	–	–	–
Testing samples	80	200	200	200	200	20
Image size(PAN image)	128 * 128	128 * 128	128 * 128	128 * 128	1024 * 1024	512 * 512

Table 2

Quantitative comparison on three datasets. Best results highlighted in red. The second best results highlighted in blue. $\uparrow$ indicates that the larger the value, the better the performance, and $\downarrow$ indicates that the smaller the value, the better the performance.

Method	WorldView-II					Gaofen-2					WorldView-III				
	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	Q2n $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	Q2n $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	Q2n $\uparrow$
SFIM	34.1297	0.8975	0.0439	2.3449	0.7843	36.9060	0.8882	0.0318	1.7398	0.6856	21.8212	0.5457	0.1208	8.9730	0.6301
Brovey	35.8646	0.9216	0.0403	1.8238	0.7626	37.2260	0.9034	0.0309	1.6736	0.4327	22.5060	0.5466	0.1159	8.2331	0.6188
IHS	35.6376	0.9176	0.0423	1.8774	0.6921	37.7974	0.9026	0.0218	1.3720	0.5265	22.5608	0.5470	0.1217	8.2433	0.6060
GS	35.2962	0.9027	0.0461	2.0278	0.8108	38.1754	0.9100	0.0243	1.5336	0.6568	22.5579	0.5354	0.1266	8.3616	0.6260
GFPCA	34.5581	0.9038	0.0488	2.1411	0.7068	37.9443	0.9204	0.0314	1.5604	0.6145	22.3344	0.4826	0.1294	8.3964	0.5027
BDSD	16.2650	0.7867	0.0346	16.5617	0.2294	10.5904	0.7438	0.0186	26.800	0.0460	21.7590	0.4867	0.1061	9.0833	0.8515
SFNLN	33.8636	0.8957	0.0481	2.2415	0.8300	36.2367	0.8850	0.0363	1.8611	0.7379	21.7647	0.5636	0.1494	9.0962	0.6727
LRTCFPan	34.7756	0.9112	0.0426	2.0075	0.8602	36.9253	0.8946	0.0332	1.7060	0.7414	22.1574	0.5735	0.1380	8.6796	0.6847
PanNet	40.8176	0.9626	0.0257	1.0557	0.8975	43.0659	0.9685	0.0178	0.8577	0.8284	29.6840	0.9072	0.0851	3.4263	0.9085
MSDCNN	41.3355	0.9664	0.0242	0.9940	0.9027	45.6847	0.9827	0.0135	0.6389	0.8977	30.3038	0.9184	0.0782	3.1884	0.9328
SRPPNN	41.4538	0.9679	0.0233	0.9899	0.9113	47.1998	0.9877	0.0106	0.5586	0.8963	30.4346	0.9202	0.0770	3.1553	0.9373
INNformer	41.6903	0.9704	0.0227	0.9514	0.9150	47.3528	0.9893	0.0102	0.5479	0.8989	30.5365	0.9225	0.0747	3.1142	0.9344
SFNet	41.7244	0.9725	0.0220	0.9506	0.9188	47.4712	0.9901	0.0102	0.5479	0.9008	30.5901	0.9236	0.0741	3.0798	0.9343
MSDDN	41.8435	0.9711	0.0222	0.9478	0.9165	47.4101	0.9895	0.0101	0.5414	0.9008	30.8645	0.9258	0.0757	2.9581	0.9367
PanFlowNet	41.8548	0.9712	0.0224	0.9335	0.9150	47.2533	0.9884	0.0103	0.5512	0.8965	30.4873	0.9221	0.0751	2.9531	0.9346
Ours	42.2354	0.9729	0.0212	0.8975	0.9229	47.6453	0.9894	0.0103	0.5286	0.9109	31.1740	0.9302	0.0698	2.8910	0.9422

Table 3

Evaluation of the proposed method on real-world full-resolution scenes from the WorldView-II dataset. Best results highlighted in red. The second best results highlighted in blue. $\uparrow$ indicates that the larger the value, the better the performance, and $\downarrow$ indicates that the smaller the value, the better the performance.

Metric	SFIM	Brovey	IHS	GFPCA	LRTCFPan	MSDCNN	SRPPNN	INNformer	SFNet	PanFlowNet	Ours
$D_s \downarrow$	0.1403	0.1026	0.1110	0.1139	0.1094	0.1063	0.0998	0.0995	0.1034	0.0966	0.0966
$D_s \downarrow$	0.1320	0.1409	0.1556	0.1535	0.1163	0.1443	0.1637	0.1305	0.1305	0.1274	0.1272
QNR $\uparrow$	0.7826	0.7728	0.7527	0.7532	0.7898	0.7683	0.7548	0.7858	0.7827	0.7910	0.7911

learning methods such as PanNet [27], MSDCNN [8], SRPPNN [28], INNformer [14], SFNet [9], MSDDN [10], and PanFlowNet [12]. The chosen evaluation metrics encompass PSNR, SSIM, SAM (spectral angle mapper) [6], ERGAS (relative global error in synthesis) [46], Q2n [47] and Non-reference metrics including D_s , D_λ and QNR.

4.2. Implement details

Utilizing the PyTorch framework, our code implementation and training procedures are executed on an Nvidia V100 GPU. The model features are configured with $N = 32$ channels. We use eight blocks of Mamba for feature extraction and five blocks of Mamba for cross modal feature fusion.

We initialize the learning rate at $5e-4$, employing a cosine decay scheduling strategy. After 500 epochs, the learning rate diminishes to $5e-8$. Optimization is carried out using the Adam optimizer, with gradient clipping set to 4. Given variations in data sizes, we set the number of training epochs to 200 for the WorldView-II dataset and 500 for the Gaofen-2 and WorldView-III datasets.

4.3. Comparison with state of arts methods

4.3.1. Evaluation on reduced-resolution scene

In our comparative analysis depicted in Table 2, we benchmarked our proposed method against state-of-the-art methods in the field. The results highlight significant improvement achieved by our proposed network structure, outperforming other methods across various evaluation metrics. Notably, on both the WorldView-II and WorldView-III

datasets, our method demonstrated noteworthy improvements in PSNR metrics, with enhancements of 0.38 and 0.29, respectively. This indicates a closer alignment of our results with the ground truth. Similar trends are observed in the SSIM indicator, while the SAM indicator signifies spectral similarity. Our spectral similarity on WV2 surpasses state-of-the-art methods, with comparable outcomes on the WV3 and GF2 dataset. The ERGAS indicator validate the overall superior performance of our method across each spectral band, substantiating its efficacy.

Quantitative results of the model are illustrated in Figs. 2–4. Representative samples were chosen for visualization, and the last row of each image depicts the mean square error map of the pan-sharpened result in comparison to the ground truth. Brighter regions indicate greater discrepancies. To clarify, we generated the error maps by first calculating the error as $E = |GT - Y|$ where GT is the ground truth image and Y represents the fusion result. We then normalize E to a range of (0,1) and apply a color map, where lower values are shown in blue and higher values in yellow. Our method consistently yields the smallest errors, underscoring its proximity to the ground truth. Furthermore, upon pan-sharpened the results, our method exhibits superior performance in extracting high-frequency information and preserving spectral details, leading to clearer texture.

4.3.2. Evaluation on full-resolution scene

To validate our model's performance in real-world settings, we conducted evaluations using the full-resolution WV2 (FWV2), full-resolution GF2 (FGF2) and full-resolution WV3 (FWV3) datasets. This dataset was utilized in its original form without downsampling, presenting a scenario with consistent real-world degradation. Owing to the

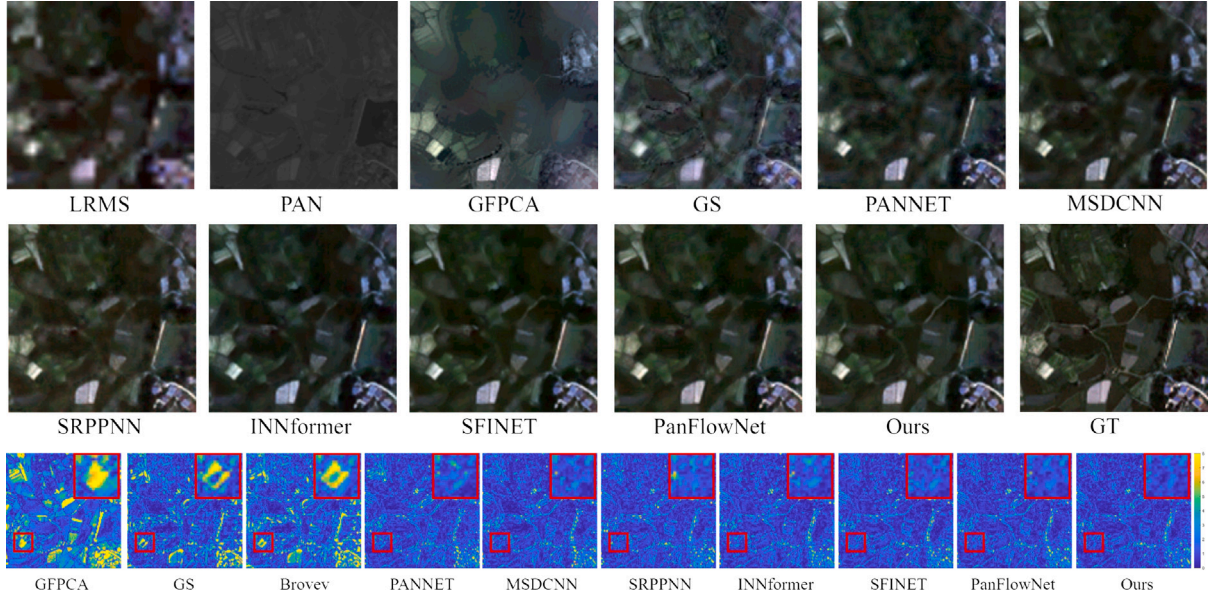


Fig. 2. The result of our approach was compared against nine other methods on WorldView-II dataset.

Table 4

Evaluation of the proposed method on real-world full-resolution scenes from the Gaofen2 dataset. Best results highlighted in red. The second best results highlighted in blue. $\uparrow$ indicates that the larger the value, the better the performance, and $\downarrow$ indicates that the smaller the value, the better the performance.

Metric	SFIM	Brovey	IHS	GFPCA	LRTCFPan	MSDCNN	SRPPNN	INNformer	SFINet	PanFlowNet	Ours
$D_\lambda \downarrow$	0.0822	0.1378	0.0770	0.0914	0.1170	0.0767	0.0998	0.0693	0.0681	0.0665	0.0652
$D_S \downarrow$	0.1087	0.2605	0.2985	0.1635	0.2024	0.1151	0.1162	0.1138	0.1119	0.1113	0.1129
QNR $\uparrow$	0.8214	0.6390	0.6485	0.7615	0.7063	0.8253	0.8173	0.8247	0.8460	0.8257	0.8306

Table 5

Evaluation of the proposed method on real-world full-resolution scenes from the WorldView-III dataset. Best results highlighted in red. The second best results highlighted in blue. $\uparrow$ indicates that the larger the value, the better the performance, and $\downarrow$ indicates that the smaller the value, the better the performance.

Metric	SFIM	Brovey	IHS	GFPCA	LRTCFPan	MSDCNN	SRPPNN	INNformer	SFINet	PanFlowNet	Ours
$D_\lambda \downarrow$	0.0107	0.0252	0.0340	0.0230	0.0631	0.0801	0.0583	0.0944	0.0507	0.0924	0.0436
$D_S \downarrow$	0.0472	0.0844	0.0955	0.0806	0.0257	0.0607	0.0559	0.0686	0.0504	0.0760	0.0444
QNR $\uparrow$	0.9426	0.8926	0.8739	0.8984	0.9129	0.8638	0.8890	0.8431	0.9016	0.8388	0.9141

absence of ground truth, we employed non-reference quality metrics — namely, D_s , D_λ , and QNR — for assessing model performance. To validate our model's performance in real-world settings, we conducted evaluations using the full-resolution WV2, full-resolution GF2 and full resolution WV3 [48] datasets. This dataset was utilized in its original form without downsampling, presenting a scenario with consistent real-world degradation. Owing to the absence of ground truth, we employed non-reference quality metrics — namely, D_s , D_λ , and QNR — for assessing model performance.

As delineated in Table 3, our approach outperforms the current State of the Art (SOTA) across all three evaluated metrics, thereby affirming our model's robust generalizability in real-world contexts. In the full-resolution GF2 and WV3 datasets, our model also achieves the second-best results, as shown in Tables 4, and 5. Given the limited sample number of our dataset, we selected the full-resolution WV3 dataset from Deng et al. [48] to evaluate model performance. In these datasets, most deep learning methods perform worse than traditional ones in the D_λ term due to significant spectral differences.

Additionally, we have provided visual results of our method applied to the three full resolution datasets in the Figs. 5–8, showcasing its superior image fusion capabilities.

4.4. Ablation study

To comprehensively assess the contribution of each module, we conducted ablation experiments. We conducted training and reported experimental results on the WorldView-II and WorldView-III datasets, maintaining consistent experimental configurations across all ablation experiments. Our ablation experiment consists of two main parts. The first part compare the network's core operator with prevalent operators in low-level vision. The second part validates the proposed three modules by selectively removing or substituting each module to assess its individual functionality. As can be seen from the Table 6, removing any of the proposed blocks or replacing the core operations results in a decline across most metrics. For example, replacing the Mamba blocks with standard convolution layers or omitting cross-channel information exchange results in lower PSNR, SSIM, and Q2n values. Specifically, in the WV2 dataset, replacing the Mamba block with self-attention leads to a notable decline in performance. Similarly, when deleting key blocks, such as the cross-Mamba block, there is a significant drop in PSNR, SSIM, and Q2n values. In the WV3 dataset, the removal of both the cross-Mamba block and the channel-swapping Mamba block has an even greater impact on the fusion results.

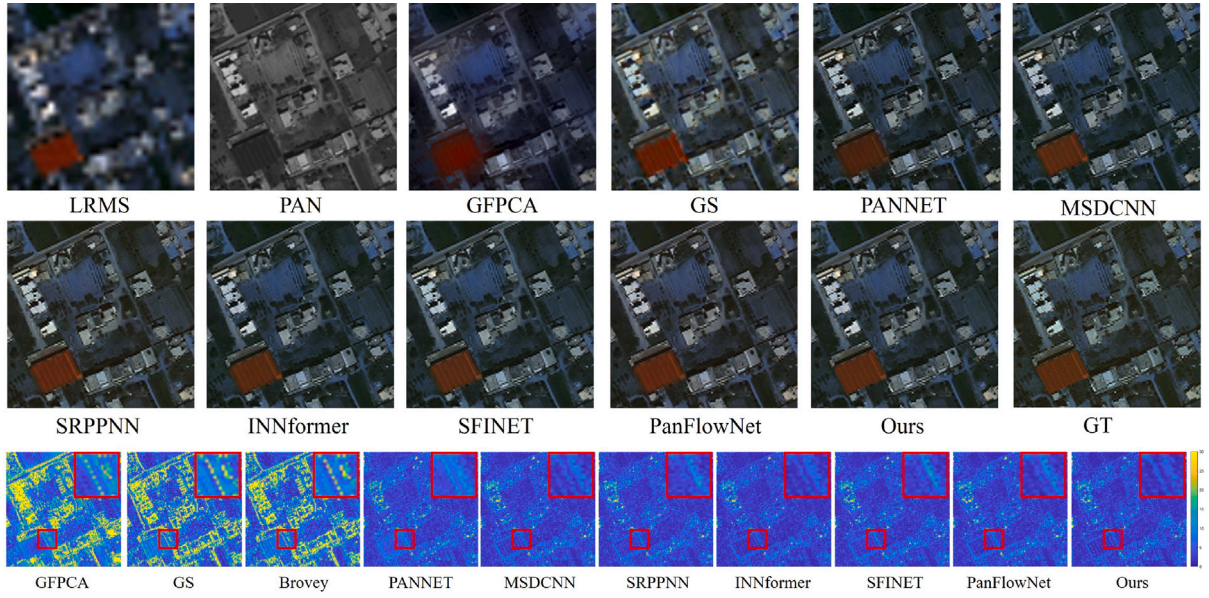


Fig. 3. The result of our approach was compared against nine other methods on WorldView-III dataset.

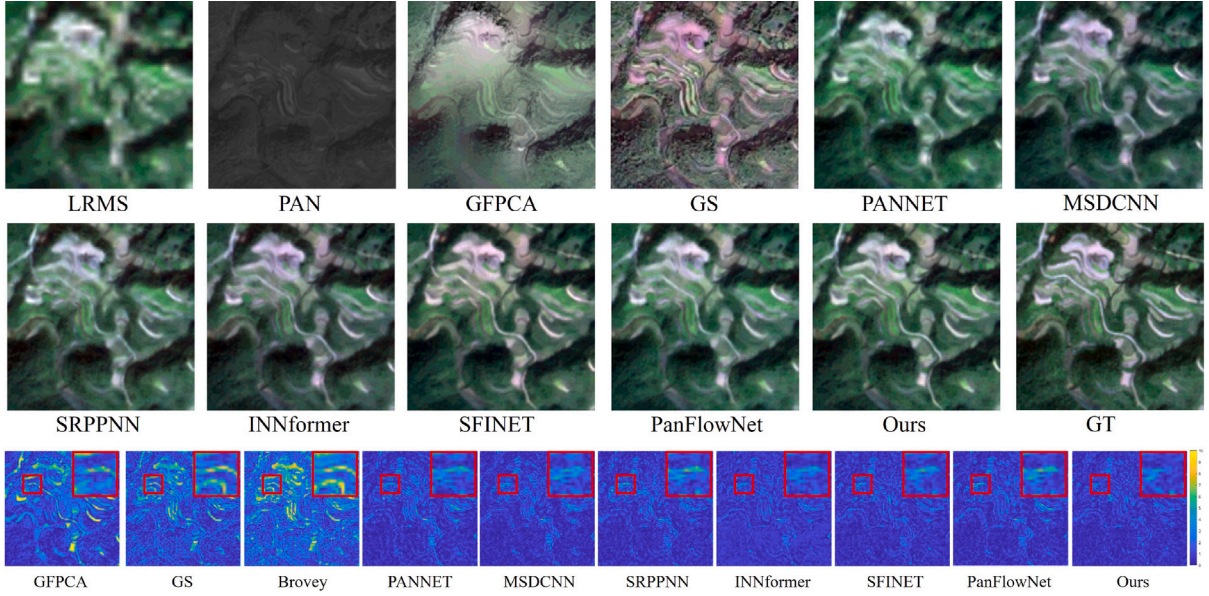


Fig. 4. The result of our approach was compared against nine other methods on Gaofen-2 dataset.

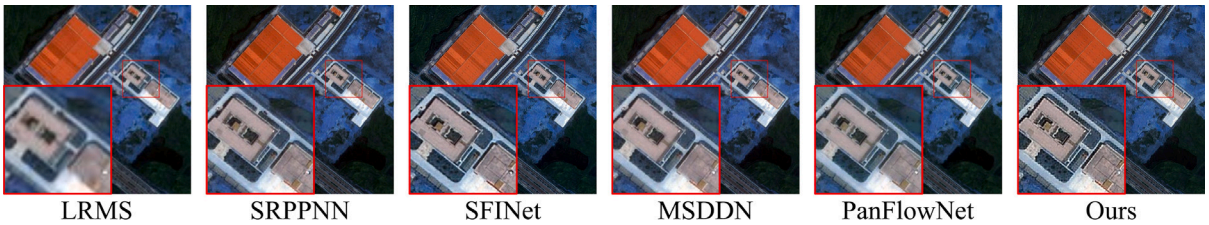


Fig. 5. The result of our approach was compared against four other methods on full-resolution WV2 dataset.

4.4.1. Effectiveness of the state space model

The ablation experiment conducted in the first part aims to validate the performance of our network's core operator, the state space model, as illustrated in rows two to four of Table 6. During this experiment, we substituted the state space model with four widely recognized operators in the image restoration domain: the convolution

operation from Hinet [49], transposed attention from restormer [19], window attention from swinir [50], and self-attention from IPT [51]. The feature extraction module was constructed directly employing the aforementioned operators. In the fusion module, Convolution layers is a hybrid of Hinet and convolution layer with 1×1 kernel, while other types utilize their respective cross-attention variants. Consistent

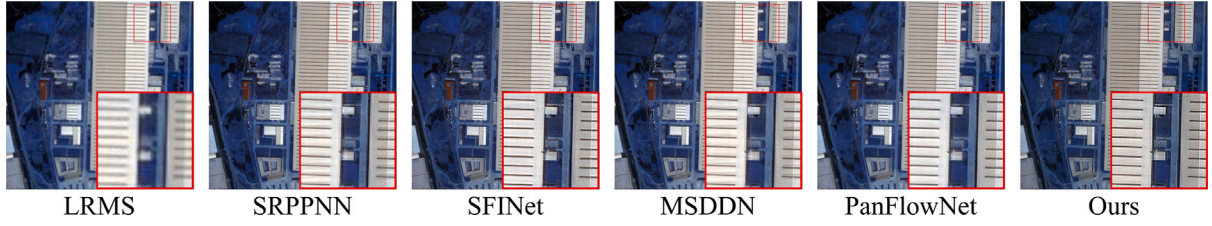


Fig. 6. The result of our approach was compared against four other methods on full-resolution WV2 dataset.

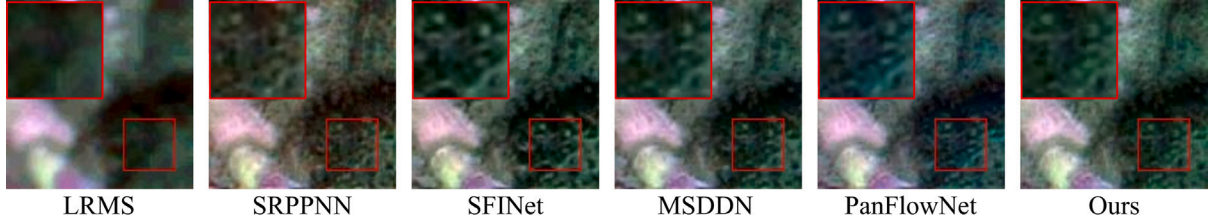


Fig. 7. The result of our approach was compared against four other methods on full-resolution GF2 dataset.

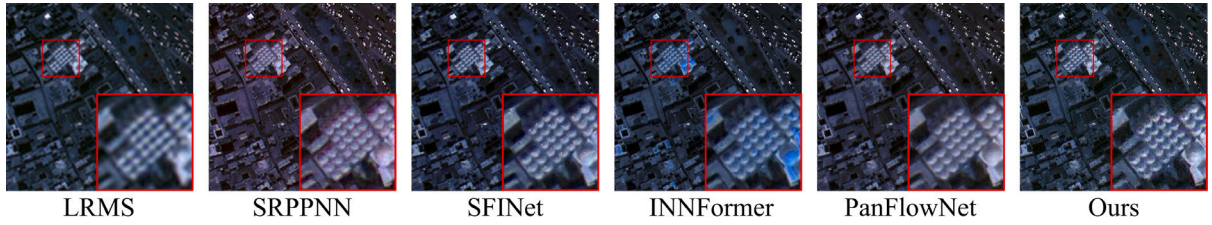


Fig. 8. The result of our approach was compared against four other methods on full-resolution WV3 dataset.

Table 6

Ablation for Pan-Mamba on Worldview-II/III datasets. The PSNR/SSIM/SAM/ERGAS/Q2n values on benchmarks are reported. “A → B” is to replace A with B. #Params are the numbers of parameters, #FLOPs are counted by the thops library with a resolution of 128×128 pixels. N and K are the number of mamba blocks and Cross Mamba blocks. Best results are highlighted in red.

Ablation	Variant	#Params [M]	#FLOPs [G]	WorldView-II	WorldView-III
Baseline	–	0.1827	3.0028	42.2354/0.9729/0.0212/0.8975/0.9229	31.1740/0.9302/0.0698/2.8910/0.9422
Core operation	Mamba->Conv	0.6235	10.1879	42.0059/0.9722/0.0214/0.9208/0.9204	31.0889/0.9295/0.0709/2.9153/0.9412
	Mamba->Transposed Attention	0.4944	8.0861	42.1568/0.9726/0.0212/0.9016/0.9212	30.9641/0.9269/0.0722/2.9592/0.9409
	Mamba->Window Attention	0.4720	7.6913	42.0980/0.9721/0.0212/0.9041/0.9198	31.0128/0.9284/0.0694/2.9445/0.9410
	Mamba->Self Attention	3.2424	14.9305	36.4403/0.9096/0.0303/1.5648/0.8622	30.2625/0.9153/0.0741/3.1970/0.9331
Main module	Mamba Block->Conv3 × 3	0.3307	5.4248	42.1659/0.9725/ 0.0211 /0.8979/0.9218	31.0710/0.9294/0.0699/2.9083/0.9410
	Channel Swapping Mamba->None	0.1827	3.0008	42.2294/0.9727/0.0212/0.8972/0.9227	30.2748/0.9163/0.0768/3.2165/0.9335
	Cross Mamba->None	0.1477	2.4532	42.0733/0.9716/0.0215/0.9109/0.9196	30.2729/0.9166/0.0767/3.2150/0.9332
Model scale	Panmamba-large(N->11;K->8)	0.2263	3.7240	42.1296/0.9724/0.0213/0.9079/0.9222	31.1538/0.9302/0.0693/2.8980/0.9419
	Panmamba-small(N->5;K->3)	0.1453	2.3983	42.2012/0.9722/ 0.0211/0.8897 /0.9225	31.0993/0.9295/0.0702/2.9127/ 0.9432
	Panmamba-tiny(N->2;K->1)	0.1089	1.7936	42.1274/0.9725/0.0214/0.9082/0.9216	30.8426/0.9257/0.0730/2.9986/0.9405

hyperparameter settings were applied across all experiments to ensure a fair comparison. Upon these experiments, it becomes evident that the mamba module's core operation, namely the state space model, boasts the fewest parameters and requires the least computational resources. Moreover, it exhibits superior performance on the WV2 dataset, as shown in Fig. 9. It is noteworthy that self-attention, while a powerful mechanism, fails to achieve optimal results on small-scale remote sensing datasets, attributable to its inherent lack of inductive bias. Transformers generally require large datasets to perform effectively. When dataset size is limited, Transformers may not perform as well as convolutional methods, which inherently capture local features.

4.4.2. Effectiveness of the Mamba block

This ablation experiment is devised to validate the efficacy of the Mamba block in feature extraction. Within our model, the Mamba block plays a crucial role in modeling long-range dependencies in features. In

this set of experiments, we substituted the Mamba block with a standard 3×3 convolution operation and relocated the flattening operation in the model post feature extraction. The results from the sixth row of Table 6 reveal a decline in model performance when the Mamba block is removed, providing evidence of the block's effectiveness.

4.4.3. Effectiveness of the channel swapping Mamba block

The second set of ablation experiments in second part is designed to validate the effectiveness of the channel swapping Mamba. Within our architecture, the channel swapping Mamba is specifically designed for shadow feature fusion, enhancing the diversity of channel features. In this experiment, we eliminated the channel swapping operation along with its associated Mamba block. As shown in the seventh row of Table 6, despite the low computational cost of this module, the removal resulted in a noticeable decline in performance, underscoring the efficacy of this module.

Table 7
Comparisons on flops and parameter numbers.

Methods	PanNet	MSDCNN	SRPPNN	INNformer	SFINet	MSDDN	PanFlowNet	Ours
Params(M)	0.0688	0.2390	1.7114	0.0706	0.0871	0.3185	0.0873	0.1827
FLOPS(G)	1.1275	3.9158	21.1059	1.3079	1.2558	2.5085	5.7126	3.0088

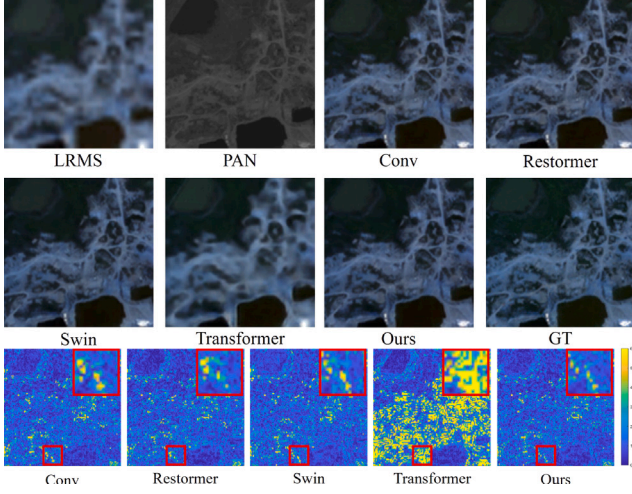


Fig. 9. Visualization of Pan-sharpening results for different core operations on WV2 datasets.

4.4.4. Effectiveness of the cross modal Mamba block

The third set of ablation experiments in second part is conducted to substantiate the effectiveness of our pivotal fusion module, the cross-modal Mamba block. This module is designed to conduct deep fusion on LRMS and PAN features, reducing redundant features through gating mechanisms. In this experiment, we directly remove the cross modal Mamba block and exclusively employed the Channel Swapping Mamba for feature fusion. The results depicted in the Table 6 showcase a substantial decline in the model's performance following the removal of the cross-modal Mamba block.

4.4.5. Comparison of model scale

To evaluate the effectiveness of our model architecture, we introduced three additional variants: Pan-Mamba-Large, Pan-Mamba-Small, and Pan-Mamba-Tiny, as detailed in Table 6. Contrary to expectations, scaling up the model to the large variant did not yield significant improvements. In fact, its slower convergence rate resulted in inferior performance metrics compared to the baseline under identical training settings. Conversely, the small and tiny versions exhibited a notable decline in performance on the WV3 and WV2 datasets.

4.4.6. Comparison of effective receptive field

To assess the effective receptive field (ERF) of our proposed methods, we present a visualization of the ERF map. We follow the methods [52] to generate Effective Receptive field. To be specific, we computed the gradient of the model's output center pixel with respect to the input image. By applying backpropagation, we identified which input pixels most significantly influenced the output. We then applied a ReLU activation to retain only the positive gradients, which highlights the positive contributions. These gradients were aggregated and averaged over multiple samples to ensure stability and generalization. Finally, the aggregated results were normalized and visualized using a heatmap. As illustrated in Fig. 11, we substitute the core operation in our model with various alternatives: conventional convolution, transposed attention, window attention, and self-attention. The figure clearly demonstrates that our method exhibits the largest receptive field, comparable to that of self-attention. In contrast, other core operations display limited receptive fields. Our approach successfully

achieves a balance between receptive field size and computational efficiency.

4.5. Comparison of efficiency

We conducted an efficiency analysis of our model, which comprises two main parts. The first part involves comparing the computational complexity between our method and several operators mentioned in the ablation study, while the second part focuses on comparing the computational complexity with the benchmark methods.

In Fig. 10, we assess the complexity of our method in comparison to Self-attention, transposed attention, window attention, and convolution, with the resolution of the input PAN image scales from 128×128 to 1024×1024 . The results reveal our method's superior performance, with the best flops metric. Regarding GPU memory usage and inference time, our method exhibits complexity comparable to transposed attention, significantly lower than window attention and self-attention. Notably, the convolution operator, despite having low memory requirements, hampers model performance due to its limited local receptive field. Conversely, the linear complexity of transposed attention lacks the ability to capture spatial dependencies, while Mamba blocks effectively model global spatial dependencies with linear complexity. In the second part of the comparison, detailed in the Table 7, our method is compared with the sota method. With similar parameter and computational complexity, our approach achieves best results, underscoring the capabilities of our model.

5. Conclusion

In this study, drawing inspiration from the State Space model, we introduce a novel Pan-sharpening network, termed Pan Mamba. This innovative network incorporates Mamba blocks, channel swapping Mamba blocks, and cross modal Mamba blocks. The proposed network achieves efficient global feature extraction and facilitates cross-modal information exchange with linear complexity. Notably, it outperforms state-of-the-art methods with a lightweight model on publicly available remote sensing datasets, showcasing robust spectral accuracy and adept preservation of texture information. Mamba's linear scan methods present a promising solution for image modeling, which can extend to related tasks such as remote sensing image super-resolution and segmentation. Furthermore, there is potential for enhancing Mamba's scan methods and cross-modal interaction, paving the way for further advancements in these areas.

CRediT authorship contribution statement

Xuanhua He: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Ke Cao:** Writing – review & editing. **Jie Zhang:** Writing – review & editing, Investigation, Conceptualization. **Keyu Yan:** Methodology, Investigation. **Yingying Wang:** Methodology, Investigation. **Rui Li:** Writing – review & editing, Supervision. **Chengjun Xie:** Writing – review & editing, Supervision. **Danfeng Hong:** Writing – review & editing, Supervision. **Man Zhou:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

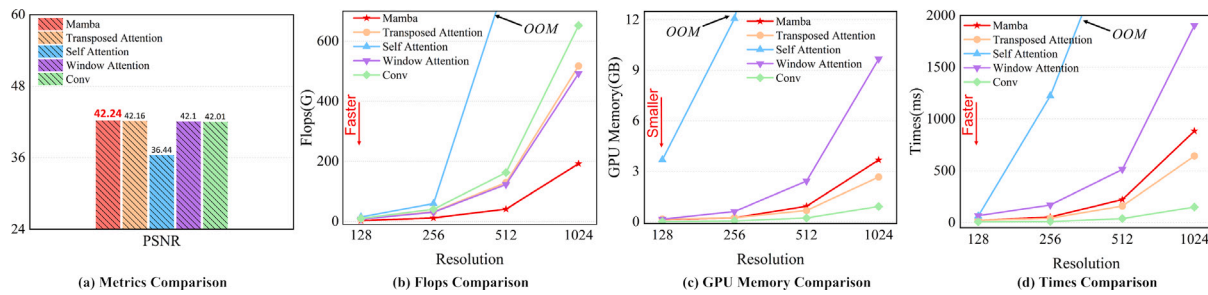


Fig. 10. Performance and efficiency comparisons between different operators in ablation section and our model.

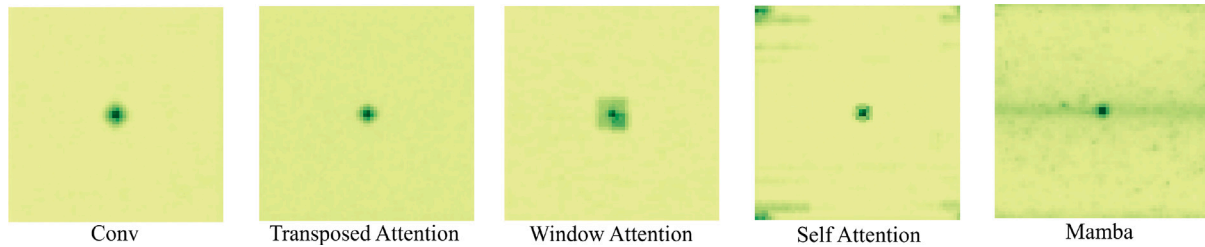


Fig. 11. Visualization of Effective Receptive Fields (ERFs) for different core operations. From left to right: Conv, Transposed Attention, Window Attention, Self-Attention, and Our Method. A larger ERF is indicated by a more extensively distributed dark area. Our method demonstrates an ERF comparable to Self-Attention, significantly larger than other operations.

Data availability

The authors do not have permission to share data.

References

- [1] J.G. Liu., Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details, *Int. J. Remote Sens.* 21 (18) (2000) 3461–3472.
- [2] A.R. Gillespie, A.B. Kahle, R.E. Walker, Color enhancement of highly correlated images. II. Channel ratio and “chromaticity” transformation techniques - ScienceDirect, *Remote Sens. Environ.* 22 (3) (1987) 343–365.
- [3] G. Vivone, Robust band-dependent spatial-detail approaches for panchromatic sharpening, *IEEE Trans. Geosci. Remote Sens.* 57 (9) (2019) 6421–6433.
- [4] X. Tian, W. Zhang, Y. Chen, Z. Wang, J. Ma, Hyperfusion: A computational approach for hyperspectral, multispectral, and panchromatic image fusion, *IEEE Trans. Geosci. Remote Sens.* 60 (2021) 1–16.
- [5] J.-L. Xiao, T.-Z. Huang, L.-J. Deng, Z.-C. Wu, X. Wu, G. Vivone, Variational pansharpening based on coefficient estimation with nonlocal regression, *IEEE Trans. Geosci. Remote Sens.* (2023).
- [6] R.H. Yuhas, A.F.H. Goetz, J.W. Boardman, Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm, 1992, URL <https://api.semanticscholar.org/CorpusID:126879175>.
- [7] G. Masi, D. Cozzolino, L. Verdoliva, G. Scarpa, Pansharpening by convolutional neural networks, *Remote Sens.* 8 (7) (2016) 594.
- [8] Q. Yuan, Y. Wei, X. Meng, H. Shen, L. Zhang, A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening, *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 11 (3) (2018) 978–989.
- [9] M. Zhou, J. Huang, K. Yan, H. Yu, X. Fu, A. Liu, X. Wei, F. Zhao, Spatial-frequency domain information integration for pan-sharpening, in: *European Conference on Computer Vision*, Springer, 2022, pp. 274–291.
- [10] X. He, K. Yan, J. Zhang, R. Li, C. Xie, M. Zhou, D. Hong, Multi-scale dual-domain guidance network for pan-sharpening, *IEEE Trans. Geosci. Remote Sens.* (2023).
- [11] H. Zhou, Q. Liu, Y. Wang, Panformer: A transformer based model for pan-sharpening, in: *2022 IEEE International Conference on Multimedia and Expo, ICME, IEEE*, 2022, pp. 1–6.
- [12] G. Yang, X. Cao, W. Xiao, M. Zhou, A. Liu, X. Chen, D. Meng, PanFlowNet: A flow-based deep network for pan-sharpening, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 16857–16867.
- [13] S. Xu, J. Zhang, Z. Zhao, K. Sun, J. Liu, C. Zhang, Deep gradient projection networks for pan-sharpening, in: *IEEE Conference on Computer Vision and Pattern Recognition*, 2021, pp. 1366–1375.
- [14] M. Zhou, J. Huang, Y. Fang, X. Fu, A. Liu, Pan-sharpening with customized transformer and invertible neural network, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36, 2022, pp. 3553–3561.
- [15] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint [arXiv:2010.11929](https://arxiv.org/abs/2010.11929).
- [16] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10012–10022.
- [17] J. Xiao, X. Fu, M. Zhou, H. Liu, Z.-J. Zha, Random shuffle transformer for image restoration, in: *International Conference on Machine Learning*, PMLR, 2023, pp. 38039–38058.
- [18] H. Yu, J. Huang, L. Li, F. Zhao, et al., Deep fractional Fourier transform, *Adv. Neural Inf. Process. Syst.* 36 (2024).
- [19] S.W. Zamir, A. Arora, S. Khan, M. Hayat, F.S. Khan, M.-H. Yang, Restormer: Efficient transformer for high-resolution image restoration, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5728–5739.
- [20] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, 2023, arXiv preprint [arXiv:2312.00752](https://arxiv.org/abs/2312.00752).
- [21] W. Liao, H. Xin, F.V. Coillie, G. Thoonen, W. Philips, Two-stage fusion of thermal hyperspectral and visible RGB image by PCA and guided filter, in: *Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing*, 2017.
- [22] R. Haydn, G.W. Dalke, J. Henkel, J.E. Bare, Application of the IHS color transform to the processing of multisensor data and image enhancement, *Natl. Acad. Sci. U. S. A.* 79 (13) (1982) 571–577.
- [23] R.A. Schowengerdt, Reconstruction of multispectral, multisensor image data using spatial frequency content, *Photogramm. Eng. Remote Sens.* 46 (10) (1980) 1325–1334.
- [24] J. Nunez, X. Otazu, O. Fors, A. Prades, V. Pala, R. Arbiol, Multiresolution-based image fusion with additive wavelet decomposition, *IEEE Trans. Geosci. Remote Sens.* 37 (3) (1999) 1204–1211.
- [25] D. Fasbender, J. Radoux, P. Bogaert, Bayesian data fusion for adaptable image pansharpening, *IEEE Trans. Geosci. Remote Sens.* 46 (6) (2008) 1847–1857.
- [26] C. Dong, C.C. Loy, K. He, X. Tang, Image super-resolution using deep convolutional networks, *IEEE Trans. Pattern Anal. Mach. Intell.* 38 (2) (2016) 295–307, <http://dx.doi.org/10.1109/TPAMI.2015.2439281>.
- [27] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, J. Paisley, PanNet: A deep network architecture for pan-sharpening, in: *IEEE International Conference on Computer Vision*, 2017, pp. 5449–5457.
- [28] J. Cai, B. Huang, Super-resolution-guided progressive pansharpening based on a deep convolutional neural network, *IEEE Trans. Geosci. Remote Sens.* 59 (6) (2021) 5206–5220, <http://dx.doi.org/10.1109/TGRS.2020.3015878>.
- [29] M. Zhou, K. Yan, J. Huang, Z. Yang, X. Fu, F. Zhao, Mutual information-driven pan-sharpening, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 1798–1808.
- [30] A. Gu, K. Goel, C. Ré, Efficiently modeling long sequences with structured state spaces, 2021, arXiv preprint [arXiv:2111.00396](https://arxiv.org/abs/2111.00396).
- [31] J.T. Smith, A. Warrington, S.W. Linderman, Simplified state space layers for sequence modeling, 2022, arXiv preprint [arXiv:2208.04933](https://arxiv.org/abs/2208.04933).

- [32] H. Mehta, A. Gupta, A. Cutkosky, B. Neyshabur, Long range language modeling via gated state spaces, 2022, arXiv preprint [arXiv:2206.13947](https://arxiv.org/abs/2206.13947).
- [33] H. Zhao, M. Zhang, W. Zhao, P. Ding, S. Huang, D. Wang, Cobra: Extending mamba to multi-modal large language model for efficient inference, 2024, arXiv preprint [arXiv:2403.14520](https://arxiv.org/abs/2403.14520).
- [34] O. Lieber, B. Lenz, H. Bata, G. Cohen, J. Osin, I. Dalmedigos, E. Safahi, S. Meirum, Y. Belinkov, S. Shalev-Shwartz, et al., Jamba: A hybrid transformer-mamba language model, 2024, arXiv preprint [arXiv:2403.19887](https://arxiv.org/abs/2403.19887).
- [35] M.A. Ahamed, Q. Cheng, Timemachine: A time series is worth 4 mambas for long-term forecasting, 2024, arXiv preprint [arXiv:2403.09898](https://arxiv.org/abs/2403.09898).
- [36] X. Jiang, C. Han, N. Mesgarani, Dual-path mamba: Short and long-term bidirectional selective structured state space models for speech separation, 2024, arXiv preprint [arXiv:2403.18257](https://arxiv.org/abs/2403.18257).
- [37] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, X. Wang, Vision mamba: Efficient visual representation learning with bidirectional state space model, 2024, arXiv preprint [arXiv:2401.09417](https://arxiv.org/abs/2401.09417).
- [38] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, Y. Liu, Vmamba: Visual state space model, 2024, arXiv preprint [arXiv:2401.10166](https://arxiv.org/abs/2401.10166).
- [39] Z. Xing, T. Ye, Y. Yang, G. Liu, L. Zhu, Segmamba: Long-range sequential modeling mamba for 3d medical image segmentation, 2024, arXiv preprint [arXiv:2401.13560](https://arxiv.org/abs/2401.13560).
- [40] J. Ma, F. Li, B. Wang, U-mamba: Enhancing long-range dependency for biomedical image segmentation, 2024, arXiv preprint [arXiv:2401.04722](https://arxiv.org/abs/2401.04722).
- [41] C.-F.R. Chen, Q. Fan, R. Panda, Crossvit: Cross-attention multi-scale vision transformer for image classification, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 357–366.
- [42] H. Zhao, O. Gallo, I. Frosio, J. Kautz, Loss functions for image restoration with neural networks, *IEEE Trans. Comput. Imaging* 3 (1) (2016) 47–57.
- [43] L. Wald, T. Ranchin, M. Mangolini, Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images, *Photogramm. Eng. Remote Sens.* 63 (1997) 691–699.
- [44] C. Laben, B. Brower, Process for enhancing the spatial resolution of multispectral imagery using pan-sharpening, 2000, US Patent 6011875A.
- [45] Z.-C. Wu, T.-Z. Huang, L.-J. Deng, J. Huang, J. Chanussot, G. Vivone, LRTCFFPan: Low-rank tensor completion based framework for pansharpening, *IEEE Trans. Image Process.* 32 (2023) 1640–1655.
- [46] L. Alparone, L. Wald, J. Chanussot, C. Thomas, P. Gamba, L.M. Bruce, Comparison of pansharpening algorithms: Outcome of the 2006 GRS-S data fusion contest, *IEEE Trans. Geosci. Remote Sens.* 45 (10) (2007) 3012–3021.
- [47] A. Garzelli, F. Nencini, Hypercomplex quality assessment of multi/hyperspectral images, *IEEE Geosci. Remote Sens. Lett.* 6 (4) (2009) 662–665.
- [48] L.-J. Deng, G. Vivone, M.E. Paoletti, G. Scarpa, J. He, Y. Zhang, J. Chanussot, A. Plaza, Machine learning in pansharpening: A benchmark, from shallow to deep networks, *IEEE Geosci. Remote Sens. Mag.* 10 (3) (2022) 279–315.
- [49] L. Chen, X. Lu, J. Zhang, X. Chu, C. Chen, Hinet: Half instance normalization network for image restoration, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 182–192.
- [50] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, R. Timofte, Swinir: Image restoration using swin transformer, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 1833–1844.
- [51] H. Chen, Y. Wang, T. Guo, C. Xu, Y. Deng, Z. Liu, S. Ma, C. Xu, C. Xu, W. Gao, Pre-trained image processing transformer, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 12299–12310.
- [52] W. Luo, Y. Li, R. Urtasun, R. Zemel, Understanding the effective receptive field in deep convolutional neural networks, *Adv. Neural Inf. Process. Syst.* 29 (2016).